

BAAZ JKBOSE



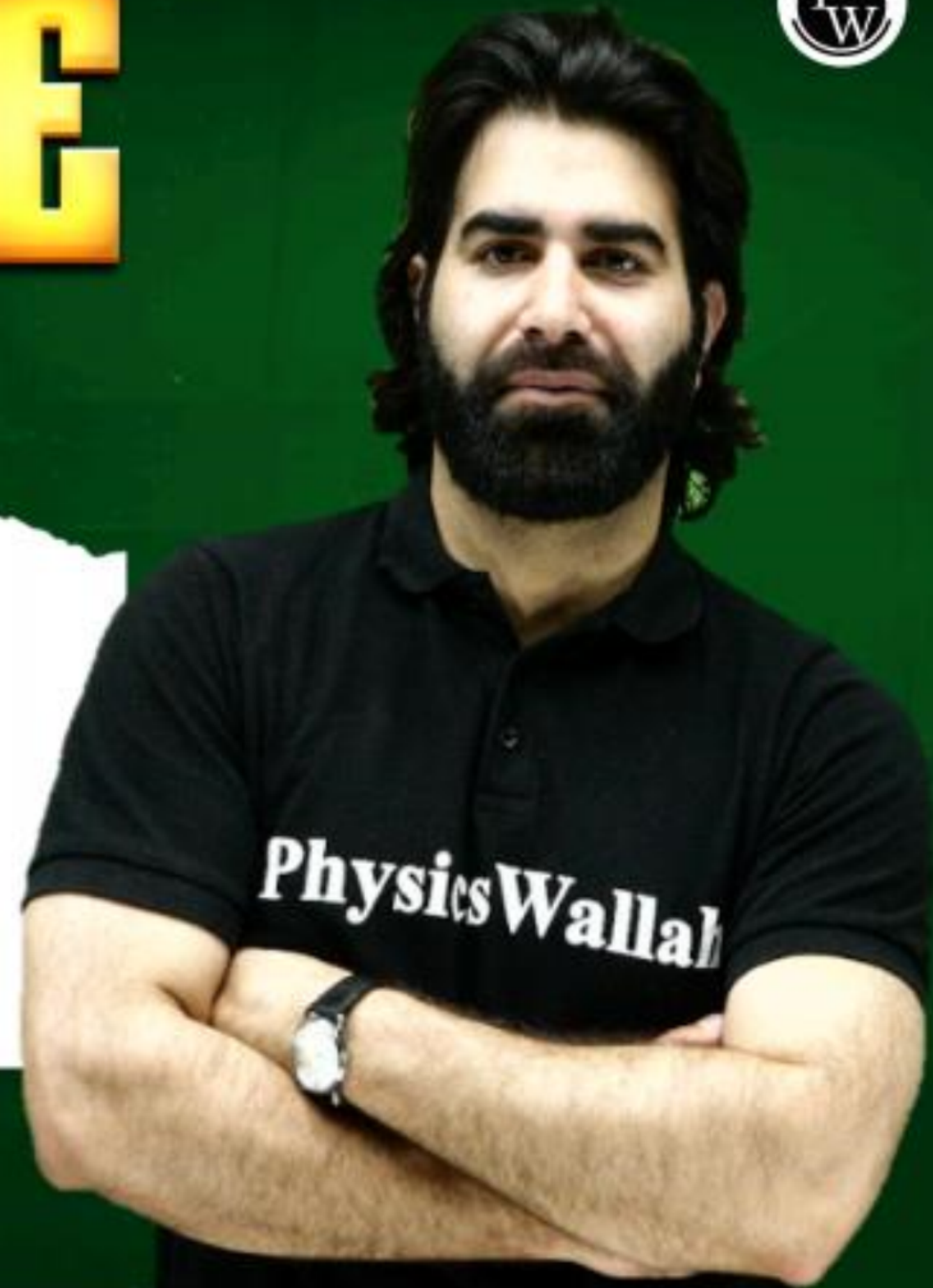
2026 CLASS-11TH

Nueral Control and
Coordination

LECTURE-3

Zoology

Dr Adil Altaf sir



Topics To Be Covered

- 1 Human Nervous System

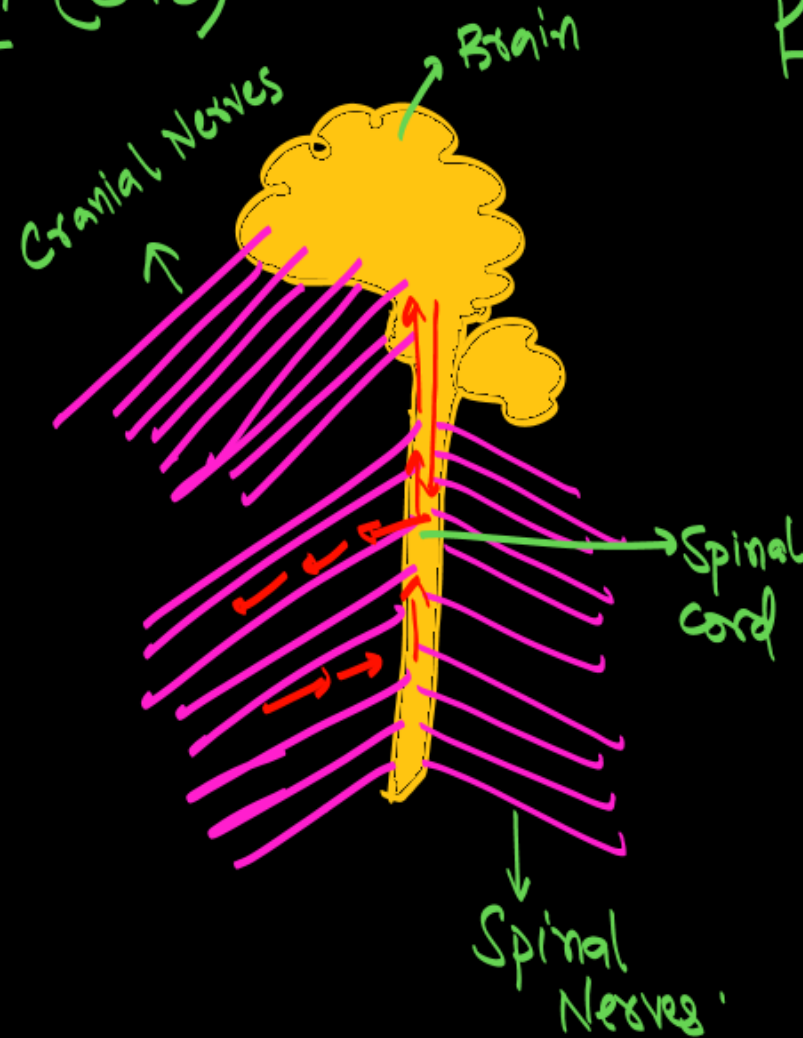
Human Nervous System (Complex type)

Nerve = Collection of Neuron fibres (Axons)



Central Nervous System (CNS)

Includes Brain & Spinal Cord.



Peripheral Nervous System (PNS)

- Cranial Nerves (12 pairs) (Mixed, Sensory, Motor)
- Spinal Nerves (31 pairs) (Mixed)
- PNS connects Body parts with CNS

PNS

Somatic Nervous System
(Voluntary)
e.g. Skeletal Muscles

Autonomic Nervous System
(Involuntary)

Parasympathetic Sympathetic

Simple type of N. System

Lower invertebrates
e.g. Hydra.

- Network of Neurons.

→ CNS

→ Coverings.

- Brain is present in Bony Box / chamber - Cranium.
- Spinal cord is present in vertebral column.

- Both Brain & Spinal are covered by 3 Meninges inside these Bones.

Human Brain contains
4- ventricles.

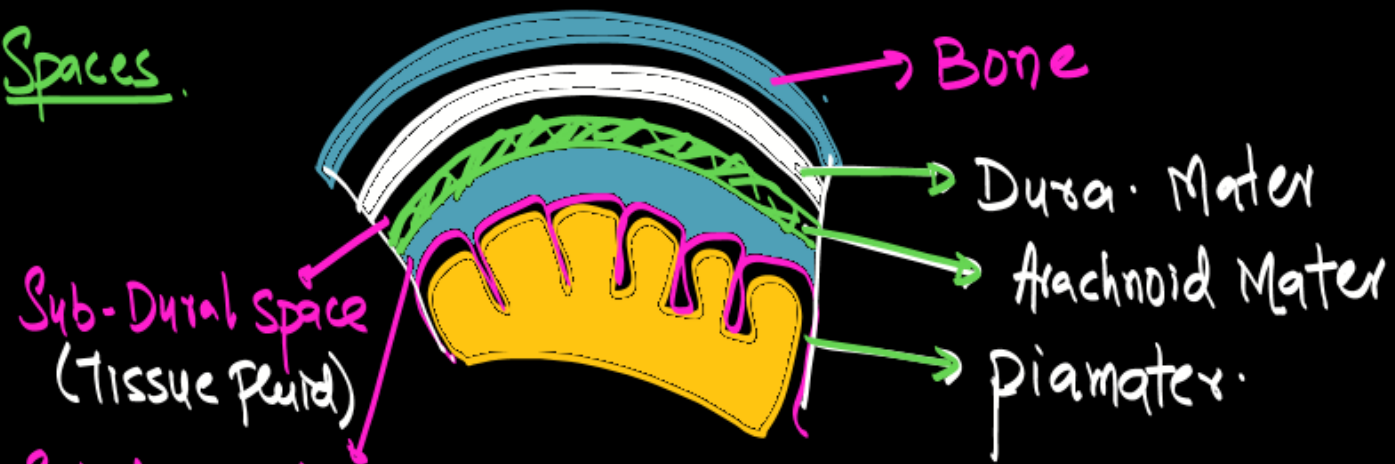
→ Meninges (Coverings)

Outer - Dura Mater
- Double layered
- Tough covering.

Arachnoid mater - Middle
- webby layers.

Piamater - Inner.
Thin, Vascular (Blood vessels).

Spaces.



Sub-Arachnoid Space.

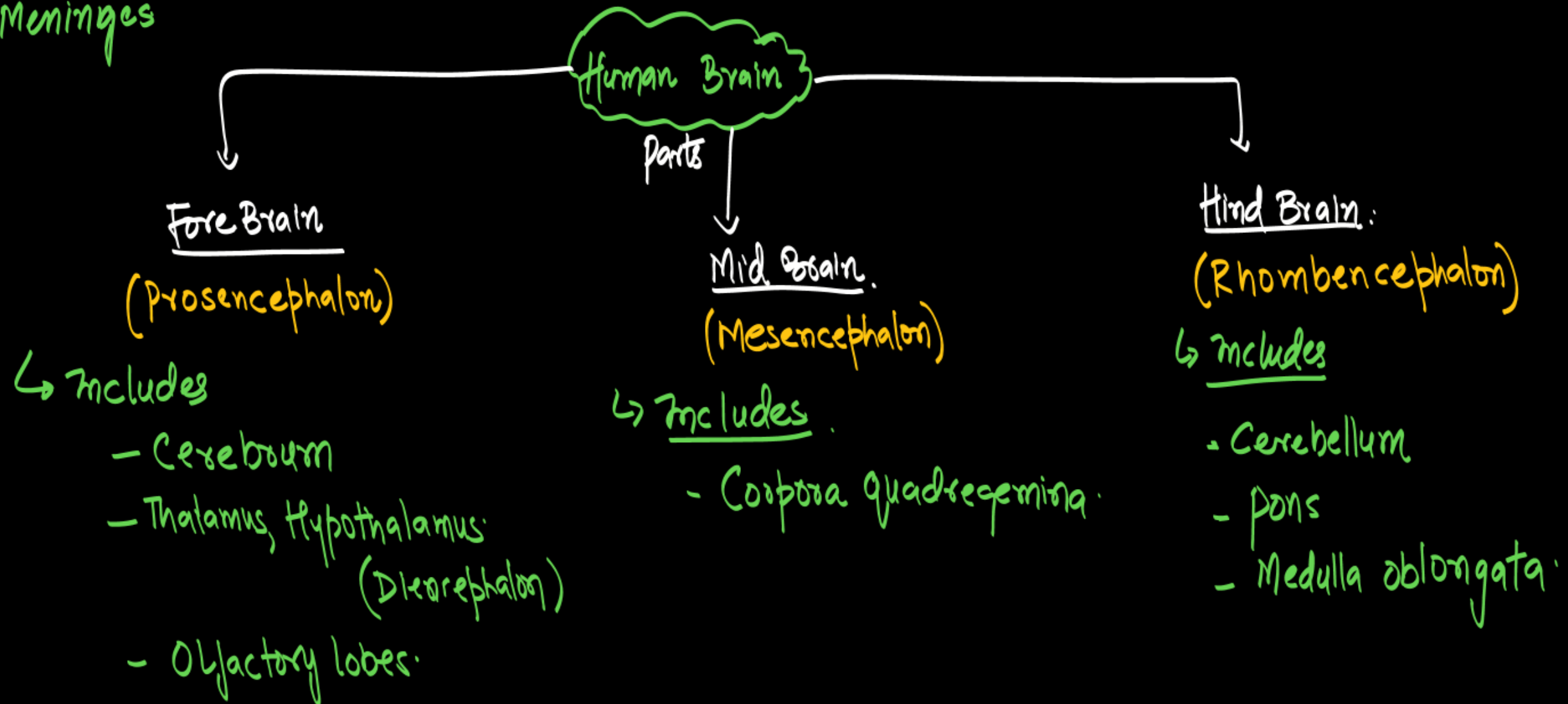
- CSF → Here as shock absorber, Also nutritious Brain Tissue.

- It is synthesized by choroid plexus in
ventricles of Brain.

Human Brain

↳ Most complex organ in the Universe

↳ Located in Cranium, Surrounded by Meninges



Cerebrum

- Largest Part of the Brain
- It has two cerebral Hemispheres (R/L)
- Surface has Sulci and gyri

Sulci - Elevations

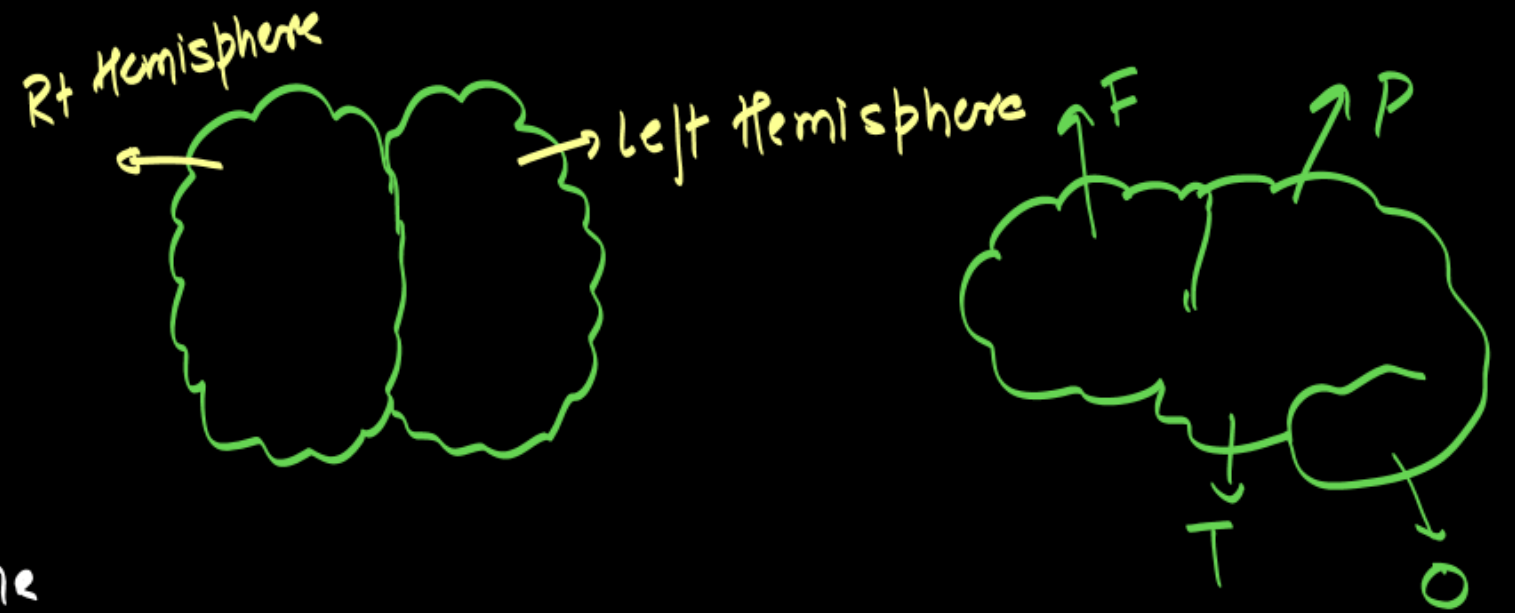
Gyri - Depressions

- These Increase Surface Area.

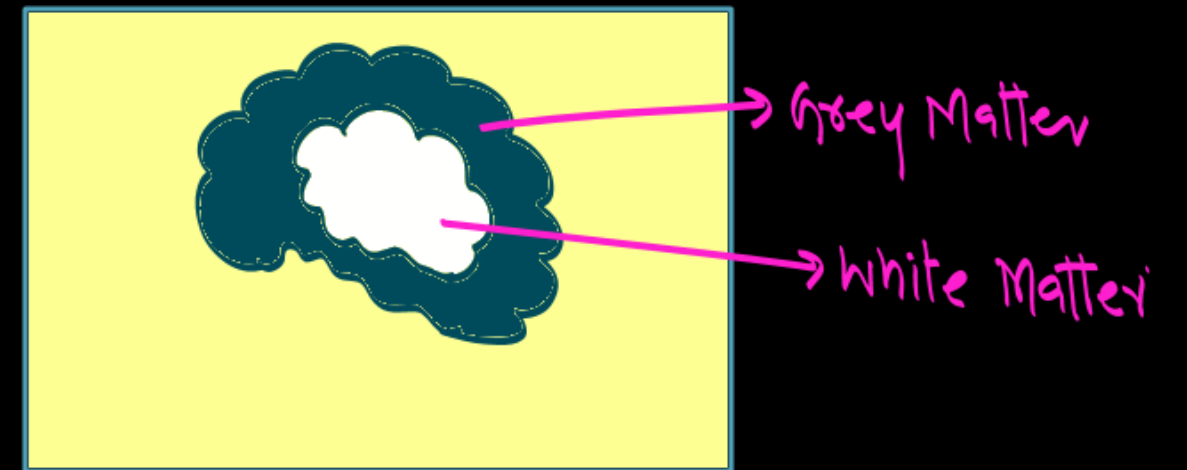
- Cerebral Hemisphere have

- Grey Matter outside (Made of Neuron Bodies).
- White Matter Inside. (Made of Myelinated Axons).

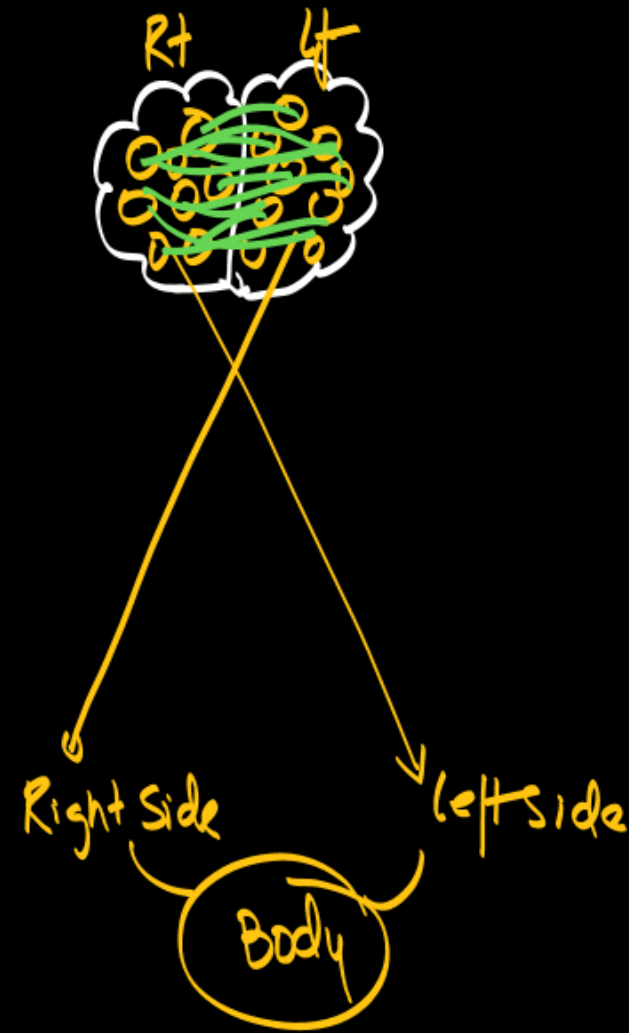
- Each Hemisphere has 4 lobes (Frontal, Parietal, Temporal, Occipital).



↓ grow



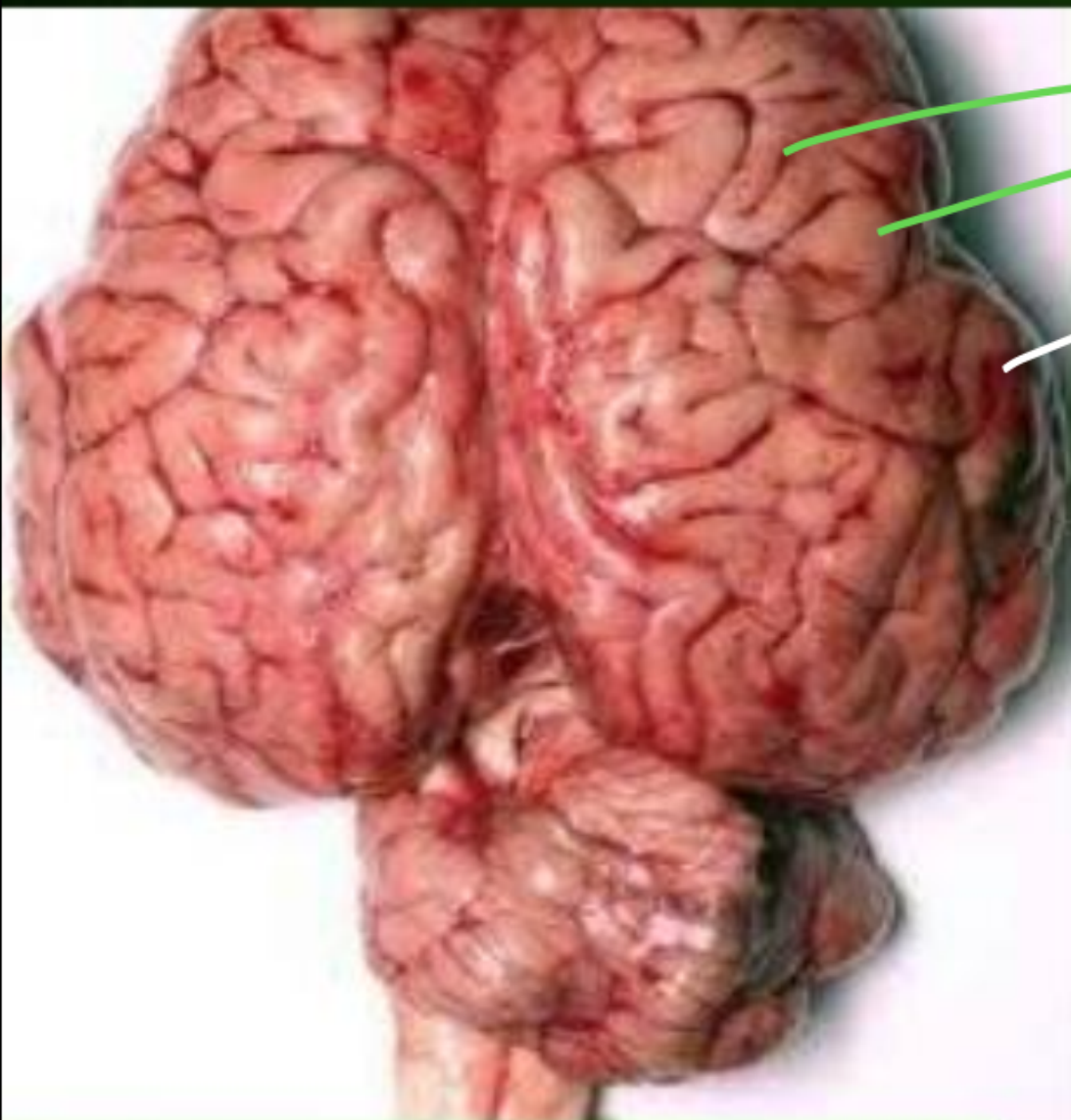
→ Rt. Cerebral Hemisphere Controls Left Side of Body and vice-versa.



→ These two Hemispheres are connected by Corpus Callosum (Network of Fibres).
[Physical, Neurological].

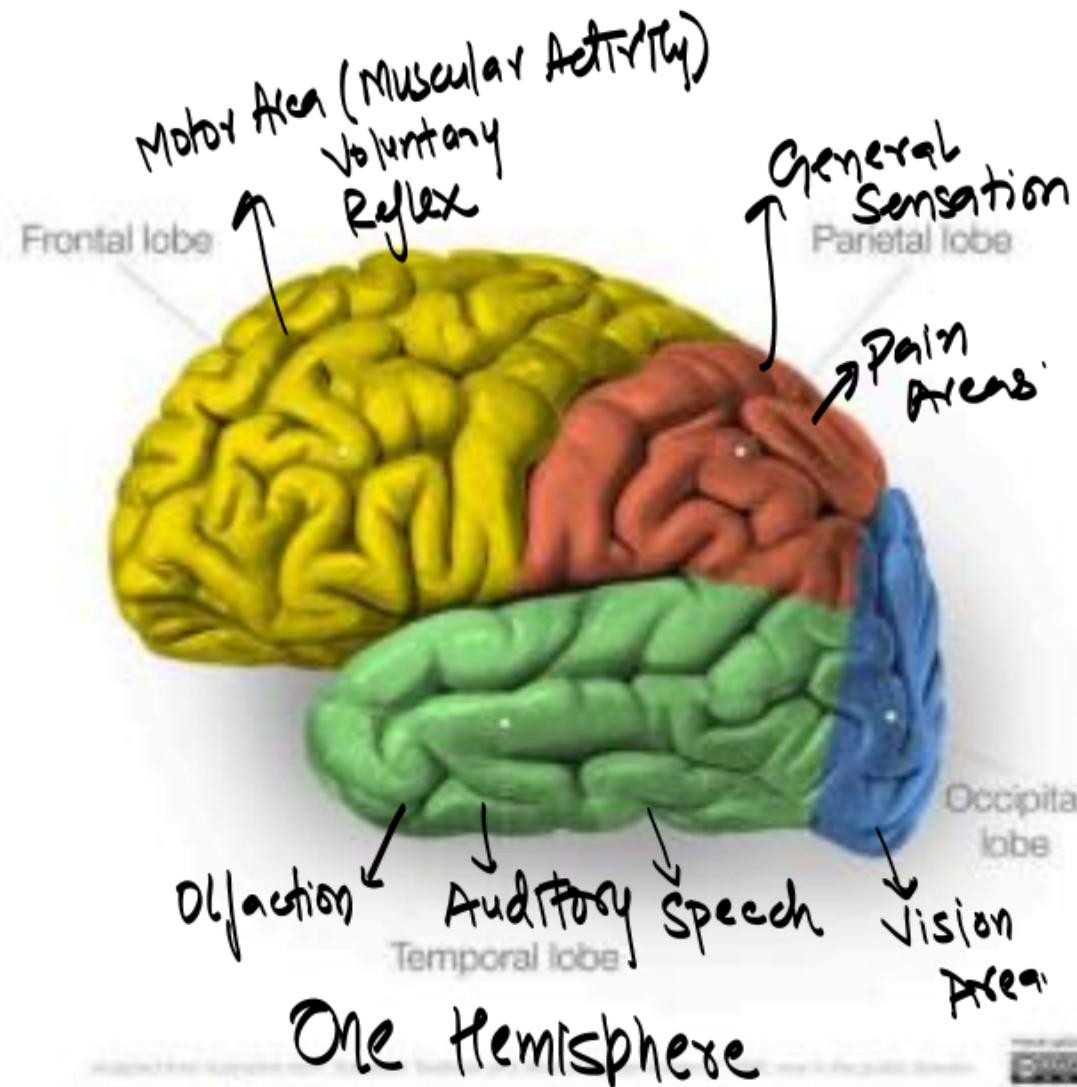
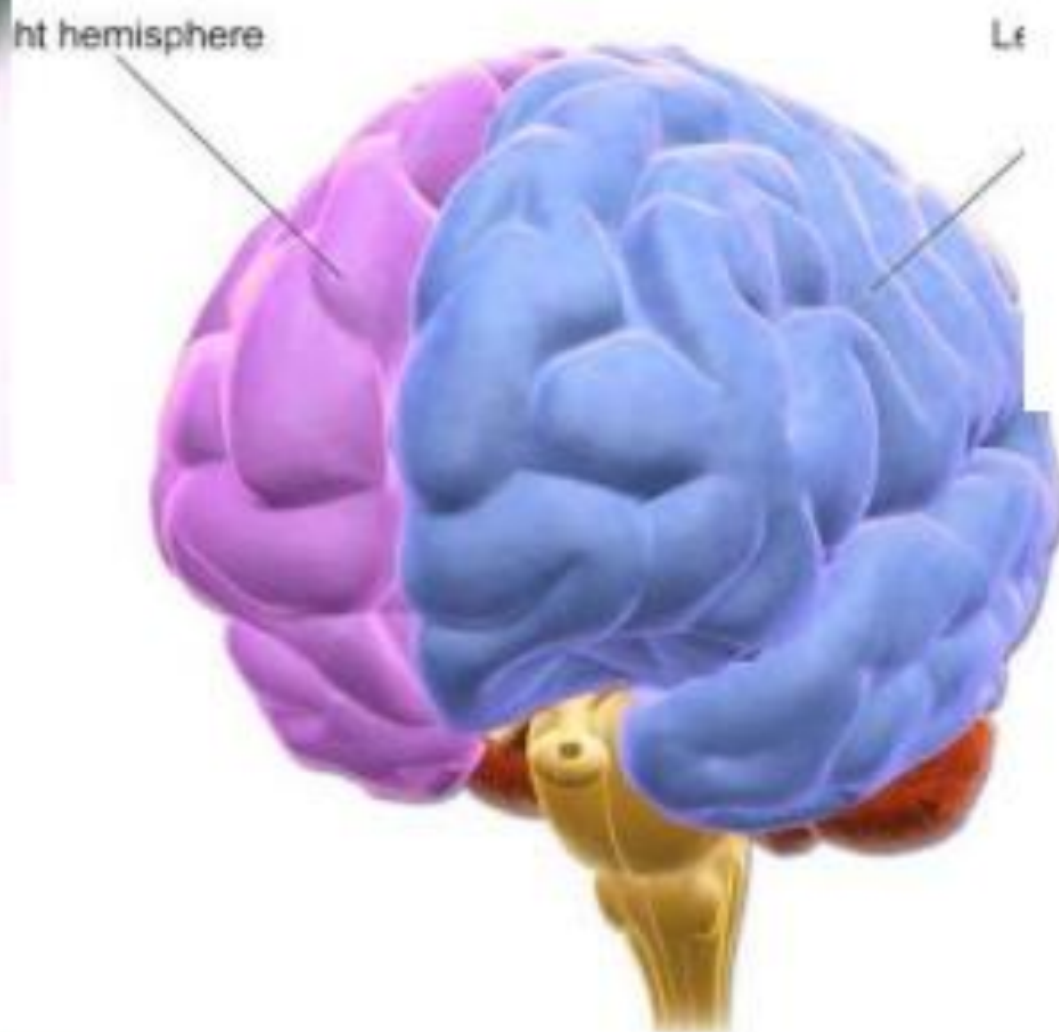
→ Cerebrum has Motor Areas (CNS → organ) e.g. Voluntary Muscle Function
Sensory Areas (organ → Brain) e.g. Visual Area, Olfactory Area.
Association Areas (Neither purely sensory, nor purely motor)

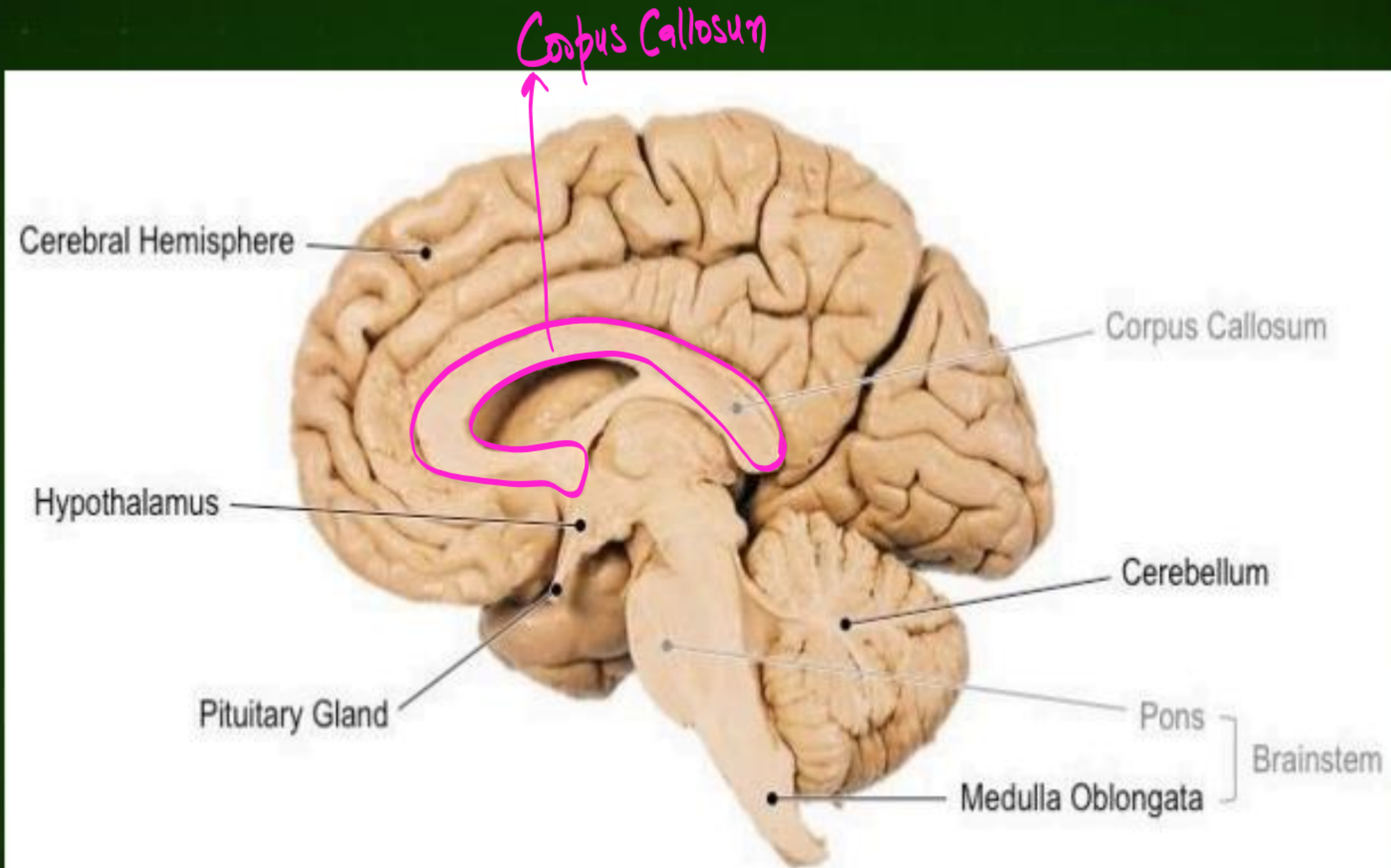
Involved in Intersensory Association, (Associating various sensory inputs correctly).
Communication.
Memory etc.



→ gyri
→ Sulci
→ Brain with pialater

Cerebral Hemispheres





Thank
You